

Design a low-power **64-bit RISC-V compute subsystem** for **mobile XR** under a **3 W, 3 nm-class LP envelope**, returning a **DSE report with evidence and rejected alternatives**.

Intent

Improve mobile XR efficiency under power, reliability, memory, and deployment constraints.

Representation /world model

XRBench traces, architecture parameters, compiler assumptions, power model, constraints.

Method role

Generate candidates, predict cost, optimize, critique assumptions, and write evidence reports.

Evidence

Pareto comparison over latency, energy, area proxy, memory traffic, QoS, and sensitivity.

Rejection authority

Checks, tool failures, QoS misses, review.

Task

Bounded DSE for a RISC-V compute subsystem; start with an accelerator and memory slice.

Environment /tool interface

Simulator or cost model plus workload harness; actions change tiling, memory, or vector width.

Feedback budget

Many cheap proxies, fewer simulations, rare high-fidelity checks before review.

Negative traces

Violates 3 W, misses real-time, exceeds bandwidth, unsupported software or fidelity failure.

Human decision

Refine, escalate fidelity, change scope, or reject.